Sensorless Position and Force Estimation for a Double-Acting Hydraulic Cylinder Using Pressure and Flow Signals

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***Abstract*—A double-acting hydraulic cylinder couples oil pressure and flow to piston position and force. We ask whether, using only pressure and flow transducers and no position sensor, position can be held within $\pm 5\,\mathrm{cm}$ and net force within $\pm 12\,\%$ of full scale across the full stroke and a 0–$50\,^{\circ}\mathrm{C}$ temperature range, under realistic sensor noise and bias. We answer with an offline study in which the cylinder, valve, supply, end-stops and controller are modelled in Hopsan, a transmission-line fluid-power simulator that supplies synthetic—and then deliberately corrupted—sensor streams to a complementary observer: position from flow-continuity velocity, re-zeroed at the end-stops, and force from the measured pressures. Across a 48-run Monte-Carlo sweep with per-run parameter mismatch, the worst-case position error is $2.58\,\mathrm{cm}$ and the operational force error $5.08\,\%$ FS, both within target. Ablations identify the end-stop *homing reset* as the load-bearing mechanism; sweeps over supply pressure, bore and hose length, a head-to-head against an extended Kalman filter, and a cross-check against a published physical experiment delimit where the scheme holds. The result is a feasibility statement for this class of actuator and a reusable plant-in-the-loop harness, not a hardware claim.**

***Index Terms*—hydraulic cylinder, sensorless estimation, state observer, Hopsan, transmission line modelling, fluid power, force estimation.**

I. Introduction

Hydraulic cylinders convert fluid power into linear force and motion and are the workhorse of heavy machinery. A double-acting cylinder is a coupled system: the oil flow rate into each chamber sets the piston velocity (and hence, by integration, position), while the pressure difference across the piston sets the net force. Knowing position and force is essential for closed-loop motion and for condition monitoring, but external position sensors (LVDTs, magnetostrictive rods, draw-wire encoders) add cost, cabling and failure modes, and are awkward to retrofit. Pressure transducers and in-line flow meters, by contrast, are cheap, robust and often already present for protection and diagnostics. This motivates *sensorless* estimation: recover position and force from pressure and flow alone.

The difficulty is that the two informative channels degrade differently. Integrating flow to get position is exact in principle but accumulates any flow-meter *bias* without bound, so a $0.5\,\%$ offset becomes metres of drift over a duty cycle. Pressure maps to force almost directly, so force is limited only by transducer noise. Temperature complicates both: over 0–$50\,^{\circ}\mathrm{C}$ the oil bulk modulus (its stiffness against compression, which sets how fast pressure builds for a given trapped-volume change), viscosity and leakage change substantially, shifting the compressibility and leakage corrections the estimator must make.

This paper is a deliberately scoped *spike*: a time-boxed feasibility study of one concrete actuator and duty cycle, with the explicit acceptance targets of $\pm 5\,\mathrm{cm}$ position and $\pm 12\,\%$ full-scale (FS) force across the stroke and the temperature range. We do not claim a new estimator class. The contributions are: (i) a physically faithful, reproducible plant-in-the-loop harness that uses Hopsan as the truth model, so the synthetic sensor data comes from a validated TLM fluid-power solver rather than a bespoke script; (ii) a complementary observer (a deliberately simple estimator that *blends* two imperfect information sources—here a fast flow-derived velocity and a slow position correction at the end-stops—rather than running a full statistical filter) whose drift is bounded by end-stop homing, with steady-gated bias calibration as a refinement; and (iii) a Monte-Carlo characterisation (many runs with randomised noise and plant parameters) across temperature, sensor-noise and parameter-mismatch seeds showing the targets are met with margin, together with an identification of which mechanism actually binds.

The idea in one paragraph (Fig. 1). The piston's speed equals the net volumetric flow into a chamber divided by the piston area; integrate that speed and you have position. The net force on the piston is just the two chamber pressures times their areas. So in principle two flow meters and two pressure sensors are enough to recover both quantities with no position sensor at all. The catch is drift: any tiny constant error in a flow meter, once integrated, grows without bound, so the position estimate slowly walks away from truth. The whole design is therefore about *pinning* that drift—which is what the hydraulic end-stops, where the piston momentarily stops and its position is known exactly, let us do for free.

Roadmap. Section III builds the plant model (and explains why a transmission-line simulator suits the stiff hydraulics); Section IV adds realistic sensor imperfections; Section V derives the estimator from the chamber equations; Section VII reports the Monte-Carlo sweep and then dissects *which* mechanism carries the result, tests robustness across supply pressure, cylinder size and hose length, compares against a full Kalman filter, and validates against a published physical experiment.

II. Background and Related Work

Two equations describe a hydraulic cylinder, and both come straight from first principles. The first is conservation of volume in each chamber: oil pumped in must either move the piston or compress the trapped fluid, so the rate of pressure rise is the net inflow (minus the flow "used up" moving the piston) divided by the chamber's compliance—this is the *continuity equation*, and the fluid bulk modulus is what converts a compressed volume into a pressure. The second is Newton's law for the piston: mass times acceleration equals the pressure-times-area force from each side minus friction and load. Classical hydraulic modelling [1], [2] and the servo-control literature [3], [4] build on exactly this pair. Reading force from the chamber pressures is standard [4]; sensorless *position* is the harder half, because of the integration-drift problem noted above.

Closest to this work, Zhou et al. [5] estimate the position of a mining-support cylinder from pressure-derived flow on a physical $400\,$L/min rig and report errors within $5\,\%$ above $10\,$MPa; we validate against their measurements in Sec. VII-G.

Our truth model is built in Hopsan, an open-source multi-domain simulator from Linköping University [6], [7]. Hopsan uses the transmission-line method (TLM) [8]: components communicate through bilateral delay lines with a one-time-step transport delay, which decouples the numerical solution of connected sub-models and gives unconditionally stable, distributed integration. This is attractive for a plant-in-the-loop study because the hydraulic stiffness (large β, small chamber volumes) that makes naive explicit integration fragile is handled natively by the solver.

III. Plant Model in Hopsan

The plant is the Hopsan model of Fig. 3, adapted from the Apache-licensed "Position Servo" example. The double-acting cylinder (`HydraulicCylinderC`) is connected through a 4/3 proportional servo valve (`Hydraulic43Valve`) to a pressure-relief-limited supply; a translational mass with hard end-stops carries the load, and a PI controller closes a position loop around a reference source. All components are standard library parts.

Crucially for realism, the pressure and flow transducers are not at the cylinder but at the *valve manifold*, with a hose run carrying oil to the cylinder— the usual layout on real machines, where instrumenting the moving cylinder is exactly what one is trying to avoid. We model each hose as a Hydraulic Volume (line compliance) followed by a laminar orifice (friction loss), giving the TLM-legal chain valve $\rightarrow$ line $\rightarrow$ cylinder; the orifice coefficient follows Hagen–Poiseuille ($\propto d^4/(\mu L)$, hence temperature dependent) and the volume is the hose bore volume. The sensors therefore read the manifold side, upstream of the line loss, so the estimator never sees the true cylinder pressure exactly (Fig. 2; the full ISO schematic with the hoses highlighted is Fig. 6)—a deliberate, quantifiable error source studied in Sec. VII-D.

A. Why a transmission-line plant: TLM versus acausal solvers

Because the entire premise of this study is that the *plant*, not the estimator, supplies ground truth, the choice of plant solver matters and is worth making explicit. The two mainstream options for fluid power are acausal, equation-based environments (Simscape Fluids, Modelica/Dymola, AMESim) and the transmission-line method (TLM) used by Hopsan. They differ in how a connected network is solved.

Acausal solvers treat the whole circuit as one differential-algebraic equation (DAE) system: each component contributes constitutive equations in terms of port efforts (pressure) and flows, and at every time step a global solver assembles and solves the coupled system implicitly, resolving the algebraic loops that connection imposes [9]. Fidelity and convenience are high, but a hydraulic circuit is numerically *stiff*: a large bulk modulus β acting on a small chamber volume V gives a pressure mode with natural frequency $\sim \sqrt{\beta A^2/(mV)}$ of many kHz, far above the mechanical motion of interest. Resolving it forces a stiff implicit integrator (e.g. `ode23t`/`ode15s`) with a global Newton solve per step, and stiff events—valve transitions, end-stop contact, cavitation—can stall the step-size controller.

TLM instead decouples the network physically. A fluid line has a real propagation delay; TLM models each connection as a bilateral delay line that carries pressure/flow "characteristics" with a one-time-step transport lag [6], [8], [10]. Because a component at time t sees its neighbours only through their state one step *earlier*, every component integrates locally and independently—there is no global system to assemble, no algebraic loop to resolve, and the coupling can never inject the spurious high-frequency feedback that destabilizes an explicit global scheme. The characteristic impedance of each delay line provides physically-motivated numerical damping, so the scheme is robust precisely at the stiff pressure node that troubles a monolithic solver. The practical consequences for a plant-in-the- loop Monte-Carlo study like ours are: (i) *stability*—the stiff β/V dynamics are handled natively rather than by shrinking a global step; (ii) *speed and scale*—local, decoupled updates parallelize across components and cores, so the dozens of multi-second runs behind every sweep here are cheap; and (iii) *determinism*—a fixed transport delay means fixed compute per step, which is also why TLM is favoured for hardware-in-the-loop.

Two honest caveats keep this in proportion. First, TLM's signature advantage, the faithful capture of pressure-wave propagation, is decisive for *long transmission lines* but marginal for a single lumped actuator, whose chamber delays are microseconds; here the win is numerical (stability, speed), not wave-fidelity. Second, a numerically robust *plant* does not make the *estimation* problem easy: the stiff velocity–pressure oscillator still has to be confronted in the observer and EKF (Sec. VII-F), independent of how the truth was generated. We use Hopsan because it gives a fast, stable, free, FMI-exportable truth model for the stiff regime we sweep; the estimator results that follow are agnostic to that choice.

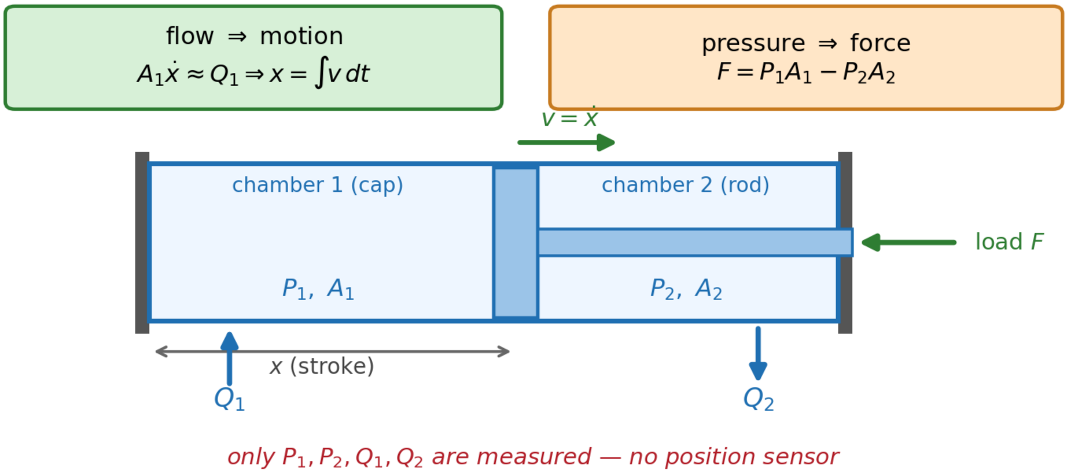

Fig. 1. The double-acting cylinder as a coupled system, from first principles. Oil flow into a chamber must move the piston (a flow Q_1 sweeps volume A_1 at speed $\dot{x}$), so integrating the flow-derived velocity gives position; the net force is the pressure-area difference $P_1A_1 - P_2A_2$. Only the four chamber signals P_1, P_2, Q_1, Q_2 are instrumented—there is no position sensor, which is the whole point of the study.

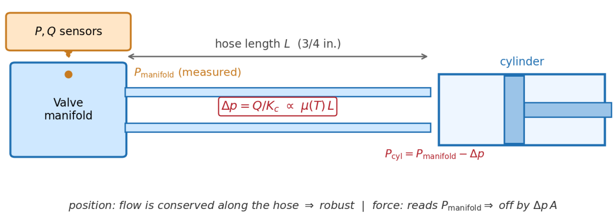

Fig. 2. Manifold-located sensing. The transducers tap the valve block; a hose of length L carries oil to the cylinder, dropping pressure by Δp. Position is unaffected (flow is conserved along the hose) but the force read-out inherits the full $\Delta p\,A$ offset, which grows with length and with colder, more viscous oil.

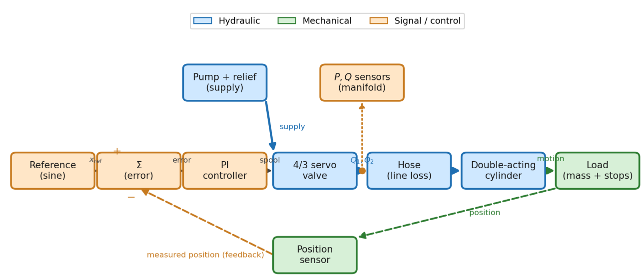

Fig. 3. Block schematic of the Hopsan plant: a PI position controller drives a 4/3 servo valve feeding a double-acting cylinder against a mass / end-stop load, a relief-limited pump supplies pressure, and a position sensor closes the loop. Blue is the hydraulic domain, green mechanical, orange signal/control. The detailed model as built in Hopsan is shown in Fig. 5.

B. Governing equations

For the cap-side (chamber 1, area A_1) and rod-side (chamber 2, area A_2) chambers, Hopsan's cylinder enforces the continuity relation

$$\dot{P}_i = \frac{\beta_e}{V_i(x)}\Big(Q_i - (\pm)A_i\dot{x} - q_{\text{leak}}\Big), \tag{1}$$

with position-dependent volumes $V_1(x) = V_{01} + A_1x$ and $V_2(x) = V_{02} + A_2(s_\ell - x)$, laminar cross-piston leakage $q_{\text{leak}} = C_{\text{leak}}(P_1 - P_2)$, and effective bulk modulus β_e. The piston obeys

$$m\ddot{x} = P_1A_1 - P_2A_2 - F_{\text{fric}}(\dot{x}) - F_{\text{load}} + F_{\text{stop}}(x, \dot{x}), \tag{2}$$

where F_{fric} combines viscous, Coulomb and stiction (static > kinetic) terms, and the valve passes a turbulent-orifice flow $Q = C_q\,w\,x_v\sqrt{2\,\Delta P/\rho}$ set by a second-order spool of finite bandwidth. Equations (1)–(2) are solved by Hopsan with the TLM scheme; we do not integrate them ourselves.

C. Temperature parameterization

Hopsan exposes β_e, C_{leak} and the viscous-friction coefficient as *constant* parameters; it has no built-in temperature dependence. We therefore cover the 0–50 °C envelope by re-parameterising the model once per run from the curves of Fig. 4: a bulk modulus that stiffens when cold, a Vogel-like viscosity that rises sharply when cold, and a leakage coefficient that scales with $1/\mu(T)$ (more leakage when hot and thin). Each Monte-Carlo run is at one steady oil temperature, so constant-per-run parameters are faithful.

D. Plant realism and model mismatch

To avoid the trap of an estimator that secretly knows the plant, the plant includes effects the observer does not model and is perturbed away from the observer's nominal parameters each run:

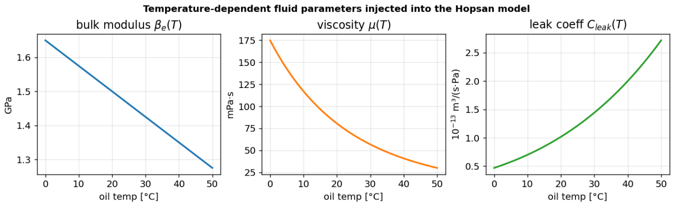

Fig. 4. Temperature-dependent fluid parameters injected into the Hopsan model per run. Hopsan itself holds these constant, so the envelope is swept by re-parameterisation.

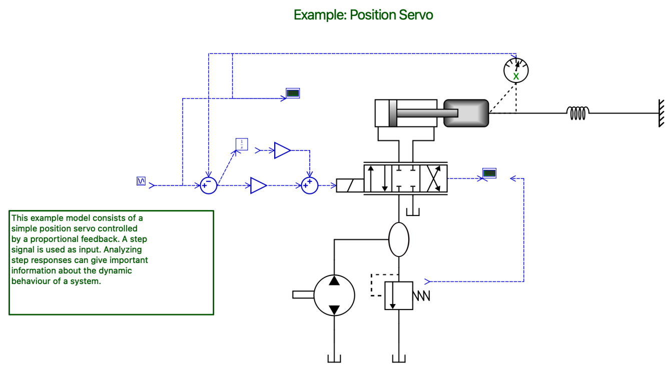

Fig. 5. The Hopsan plant model with the standard ISO fluid-power symbols: the double-acting cylinder driving a mass–spring load with the position sensor, the 4/3 servo valve, the pump and relief valve, and the blue signal/control chain (sine reference, summation, gains, integrator, scopes). This is the model that produces all synthetic sensor data; the block schematic of Fig. 3 is rendered independently from the `.hmf` for labelling.

- **Friction**: Coulomb + stiction (static 1.5× kinetic) on the piston, on top of viscous damping, producing stick–slip near reversals.
- **Valve dynamics**: a finite-bandwidth second-order spool rather than an algebraic valve.
- **Parameter mismatch**: per run the plant's bulk modulus, cross-piston leakage, viscous friction, dead volumes and areas are scaled by independent log-normal factors with one-sigma spreads of 10, 20, 20, 12 and 0.5 % respectively (representative of oil aeration/wear, seal-leakage and hose-volume uncertainty, and machined-geometry tolerance), while the observer keeps nominal values. This converts the sweep from a noise-robustness test into a genuine *model-mismatch* test.

E. Excitation and simulation

Each run is simulated in Hopsan with per-run parameters (geometry, the temperature-dependent fluid values, controller gains and the reference); the exported pressure, flow and position time series form the truth log. The reference is a sine of amplitude 1.1× stroke, so the servo sweeps the full stroke and seats both end-stops each half cycle; the seating events are what the observer uses for calibration.

IV. Synthetic Sensors

The clean Hopsan signals are corrupted by a transducer model that mimics industrial-grade hardware (Table I). Each channel applies, in order: a first-order bandwidth (pressure 500 Hz, flow 80 Hz), white noise, a per-run constant bias plus a slow bias random-walk (zero drift), a transport delay (1.5 ms), and 12-bit quantization; flow additionally gets a K-factor nonlinearity and full-scale saturation. The flow *bias* (0.5 % FS) is the term that defeats naive integration. Oil temperature is available only as a noisy reading (±1 °C). No position or velocity is measured. Fig. 7 shows a representative pressure/flow record: pressures saturate near the relief setting when the piston seats, and flows reverse with stroke direction; the faint traces are the noisy measurements the observer actually sees.

TABLE I
Cylinder and Sensor Parameters

Quantity	Value	Note
Bore / rod / stroke	100/56/1000 mm	
Moving mass	200 kg	
Supply pressure	210 bar	relief-set
Force full scale	164.9 kN	$A_1 P_s$
Pressure noise	0.5 % FS	FS = 250 bar
Pressure bias	0.25 % FS	per run
Flow noise	1 % rdg +0.2 % FS	FS = 160 L/min
Flow bias	0.5 % FS	per run
Sensor bandwidth	500 / 80 Hz	pressure / flow
Transport delay	1.5 ms	all channels
Zero drift	0.15 % FS/s	random walk
Temperature sensor	±1 °C	noise
Sampling rate	1 kHz	

V. Sensorless Observer

Position is reconstructed from the channel that actually carries it: flow. Each chamber's continuity equation (1), rearranged, gives an estimate of piston velocity from the measured flow, corrected for bias b_i, leakage and the compressibility term:

$$A_1 v_1 = (Q_1 - b_1) - (V_1/\beta_e)\dot{P}_1 - q_{\text{leak}}, \quad (3)$$
$$A_2 v_2 = (V_2/\beta_e)\dot{P}_2 - (Q_2 - b_2) - q_{\text{leak}}. \quad (4)$$

The two estimates are blended by flow magnitude (higher flow, higher signal-to-noise, more weight), and position is the time integral of the blended velocity. The fluid properties use the noisy temperature reading and the same $\beta(T)$, $\mu(T)$, $C_{\text{leak}}(T)$ curves as the plant, so the compressibility and leakage corrections track temperature with residual mismatch. Figure 8 shows the full signal flow.

The bias b_i is the binding term. We observe it at the end-stops: when the piston is seated, the true velocity is zero and, in steady state, the measured flow equals only the known leakage, so $b_1 = Q_1 - q_{\text{leak}}$ and $b_2 = Q_2 + q_{\text{leak}}$. Seating is detected from low flow plus high pressure; the bias update is *gated on steady pressure* ($\dot{P} \approx 0$) so that transient seats, where the bias identity does not hold, are rejected. The same events pin absolute position (homing), bounding integration drift. Force is read directly from the measured (filtered) pressures, $F = P_1 A_1 - P_2 A_2$. Formally this observer is a structurally-reduced

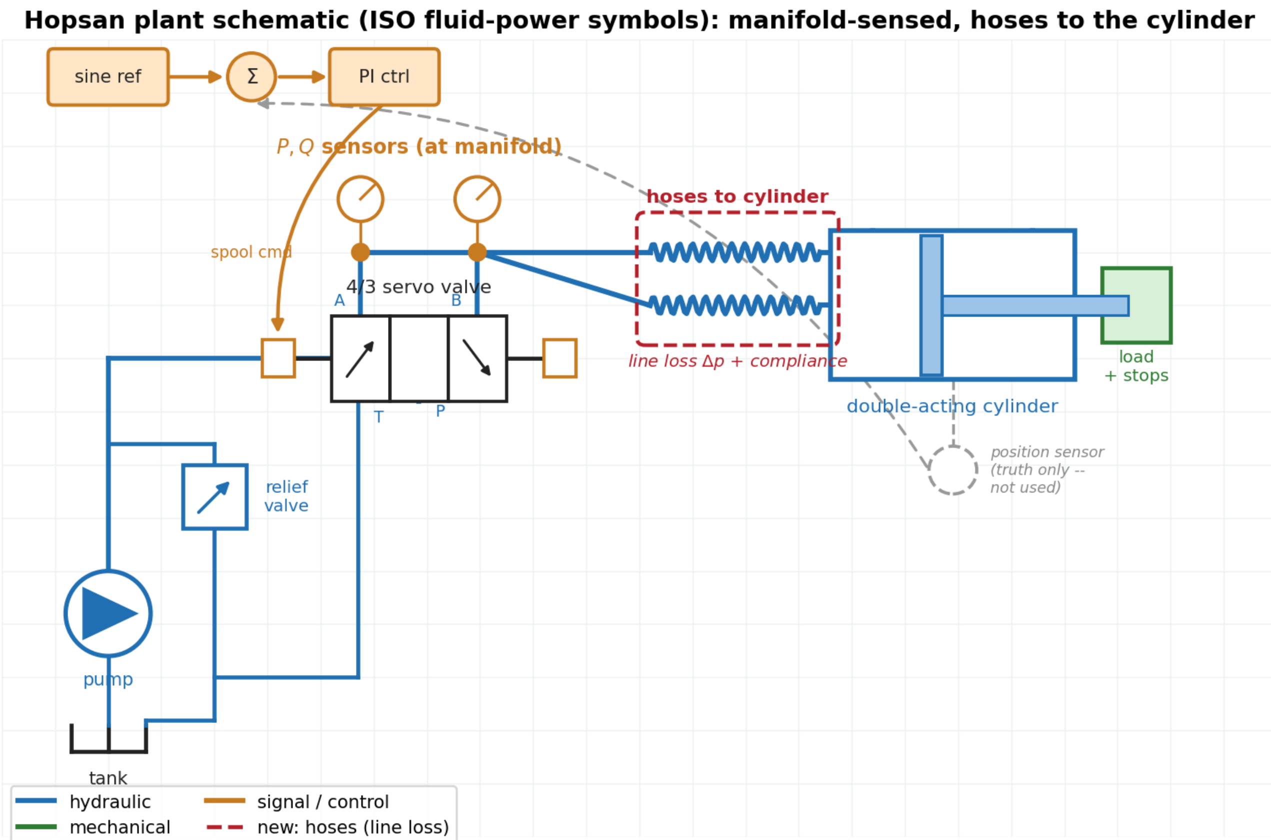

Fig. 6. Annotated ISO-1219 schematic of the same plant, with the manifold-to-cylinder hoses added this revision highlighted (red). Oil runs tank → pump → relief-limited supply → 4/3 servo valve; the P, Q transducers tap the valve manifold (ports A/B), and *flexible hoses* carry oil to the double-acting cylinder, so the sensors sit upstream of the line drop Δp. The cylinder's own position sensor (grey) is ground truth only and is never used by the estimator.

realization of an Extended Kalman Filter on the cylinder state-space model; Appendix C gives that model, the EKF recursion and Jacobians, and why the full filter is ill-conditioned for the stiff hydraulic dynamics (the reason we use the reduced form).

A seated stop gives two *distinct* corrections, easily conflated (Fig. 9). **Homing reset** uses the known stop position to set $\hat{x} \leftarrow L$ (or 0): it removes the *accumulated* error instantly, a vertical drop in the error trace. **Bias calibration** uses $v = 0$ to re-identify the flow-meter bias: it removes the *drift source*, reducing the slope at which error grows *between* seats. Reset corrects the symptom now; calibration slows future growth. Which one dominates depends on how often the piston seats, examined in Sec. VII.

VI. Experimental Setup

We sweep six temperatures $(0, 10, \ldots, 50\,°\mathrm{C})$ by eight seeds (each drawing independent sensor noise/bias and a plant-parameter mismatch), for 48 Monte-Carlo runs. Each run is a 32-second full-stroke duty cycle at 1 kHz, after a short settling window. The position metric is the maximum absolute error (± 5 cm target). For force we use the *operational* metric, the 99.5th percentile of absolute error ($\pm 12\,\%$ FS target), and separately report the transient maximum: a force-from-pressure read-out is unavoidably wrong for the few milliseconds of a sub-sample pressure reversal, so an instantaneous max is a transducer property, not an estimator one.

VII. Results

All 48 runs pass. With the realistic plant (friction, valve dynamics, parameter mismatch) and sensors (bandwidth, delay, drift, saturation), the worst-case position error across the sweep is 2.58 cm (target 5 cm) and the worst-case operational force error is $5.08\,\%$ FS at the 99.5th percentile (target $12\,\%$); mean RMS errors are 0.47 cm and $1.56\,\%$ FS. The force *max* reaches $\sim 22\,\%$ FS, but only on $< 0.1\,\%$ of samples, confined to the millisecond pressure reversals at flow direction changes where a delayed, bandwidth-limited transducer cannot follow a sub-sample pressure swing; this is a transducer limit, not an estimator one. The headline verdict is in Table II.

Fig. 10 shows a representative run at the hottest, hardest temperature. The position estimate tracks the truth through the full-stroke sweep and both end-stop seatings; the error trace stays within about 1.1 cm, with small steps at the homing events where the bias is recalibrated. The force estimate, read from pressures, sits within a few percent of full scale throughout.

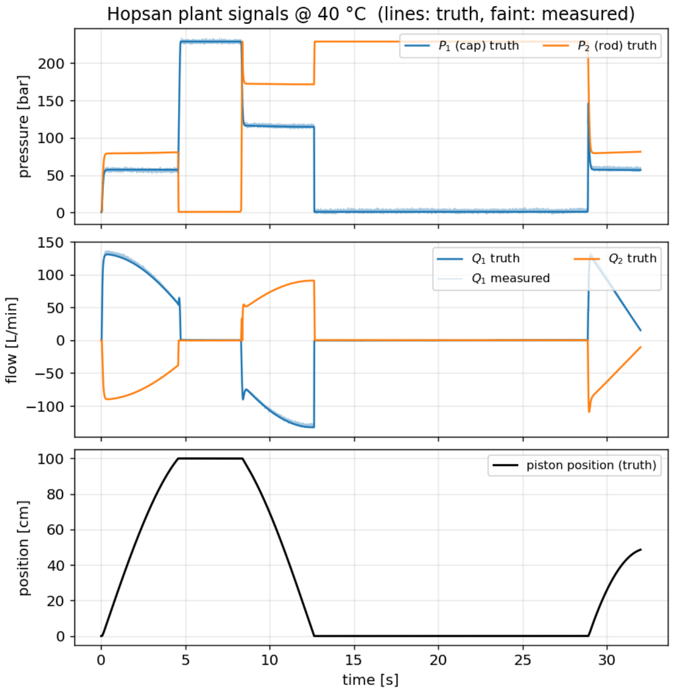

Fig. 7. Synthetic sensor streams from the Hopsan plant at 40 °C. Chamber pressures (top) climb to the relief limit at the stops; chamber flows (middle) reverse with direction; piston position (bottom) sweeps the full stroke. Faint overlays are the noisy measurements.

TABLE II
WORST-CASE ERRORS OVER THE 48-RUN SWEEP (REALISTIC PLANT + SENSORS)

Metric	Target	Worst case	Verdict
Position error	±5 cm	2.58 cm	**PASS**
Force error (99.5 pct)	±12 % FS	5.08 % FS	**PASS**
Force error (transient max)	—	$\sim$ 22 % FS	$<$ 0.1 % samples
Temperature	0–50 °C	full range	✓

Fig. 11 plots worst-case and mean error against temperature. Position error is essentially flat across the envelope: the end-stop homing reset zeros accumulated drift regardless of temperature, and the temperature-dependent leakage and compressibility corrections handle the rest. Force error is likewise flat, as expected for a pressure-derived quantity. The distribution over all 48 runs (Fig. 12) shows every run sitting well below both limits, with no temperature-correlated tail. The full ensemble of error-vs-time traces (Fig. 13) makes the spread explicit: every run's position error stays inside the ±5 cm band throughout, with the largest excursions during the mid-stroke window between homing events, while the force-error traces form a tight band near ±4 % FS.

A. Ablations: which mechanism matters

Turning observer features off one at a time (Fig. 14) shows the result is not generic. Disabling end-stop *homing* pushes the worst case to 4.9 cm (at the edge of target), so the homing reset is the primary load-bearing mechanism under this frequently-seating duty cycle. The *steady-pressure gate* is what makes bias calibration safe: removing it nearly triples the error to 7.3 cm (fail), because ungated transient seats inject compressibility flow as false bias. Notably, removing bias calibration *entirely* leaves the worst case unchanged (2.2 cm): under frequent seating the homing reset dominates, and calibration matters mainly for large bias or sparse homing. The leakage, compressibility, single-chamber and temperature variants stay near 2–2.5 cm: second-order refinements.

B. How often must the actuator home, and why?

The duty cycle seats a stop every half reference-period, so we sweep the reference frequency to vary the interval between homing events, 8 seeds per point, in three observer modes (Fig. 15). The result both gives a design rule and settles which mechanism is load-bearing. For the full observer the median error rises with interval and the ±5 cm budget holds in the median while the actuator seats a stop roughly every $\lesssim$ 30 s; the upper-quartile band reaches the limit nearer $\lesssim$ 20 s, so $\sim$ 20 s is a reasonable distributional guideline. It is not a hard guarantee: individual unlucky mismatch draws can exceed ±5 cm even at shorter intervals, so a safety-critical use would need a tail-reliability budget. Crucially, *reset-only* (calibration disabled) tracks the full observer almost exactly across the whole range, whereas *calibration-only* (homing reset disabled) is worse at every interval and diverges once seating is sparse. So the **homing reset is the load-bearing mechanism**: it zeros error from *all* sources (bias, mismatch, noise) at each seat, while bias calibration only removes the flow-meter-bias part of the between-seat slope and cannot bound dead-reckoning drift on its own. Calibration is a useful refinement (it extends the usable interval modestly), not the foundation.

C. Operating envelope: supply pressure and cylinder size

Holding the duty cycle, we sweep the two parameters a deployment varies most: supply pressure and bore. **Pressure** (Fig. 16): position error is flat from 60–280 bar, because the flow-continuity estimate barely uses pressure. Force error instead scales like $1/P_{\text{supply}}$ when the transducer full scale is fixed (a 250-bar transducer gives $\sim$ 9.5 % FS at 60 bar, near the limit), but is flat once the transducer is matched to the supply. Low-pressure operation is thus harder for force only through transducer SNR, a sizing choice, not an estimator limit. Even the deeper low-pressure physics leaves position unmoved: adding an entrained-air bulk modulus $\beta(P)$ that collapses below $\sim$ 50 bar (Fig. 17) changes position negligibly, and a naive constant-β observer matches a pressure-aware one, because the flow channel is β-insensitive.

Size (Fig. 18): across ISO bores 63–160 mm (area-similarity scaling) position error is flat at $\sim$ 2.3 cm and force at $\sim$ 2.6 % FS, *provided the flow meter is sized to the cylinder*; a fixed meter fails at both ends (saturation for large bores, noise floor for small). The estimation itself is therefore robust across the envelope, subject to two clean sensor-sizing rules: **size the pressure transducer to the supply (for force) and the flow meter to the cylinder area (for position)**.

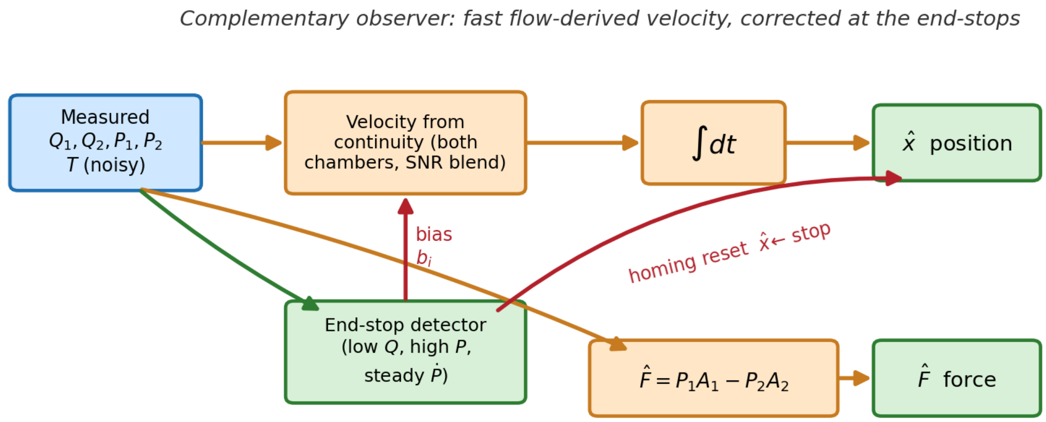

Fig. 8. The complementary observer as a signal-flow graph. The fast path (orange) turns measured flows into a velocity and integrates it to position; pressures give force directly. The end-stop detector supplies the two slow corrections (red): a *homing reset* that snaps $\hat{x}$ to the known stop, and a *bias calibration* b_i that re-zeros the flow meters. The reset is what bounds integration drift.

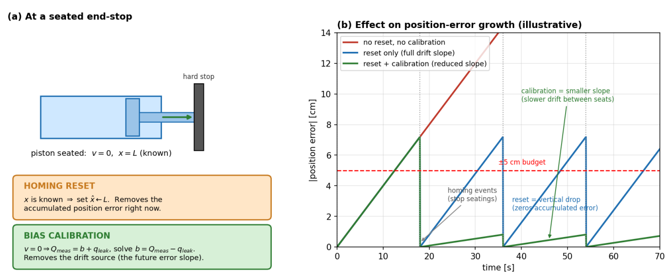

Fig. 9. The two end-stop corrections. (a) When seated, position is known (⇒ reset) and velocity is zero so measured flow is bias plus known leakage (⇒ calibrate the bias). (b) Reset zeros the accumulated error at each seat (vertical drops); calibration reduces the drift slope between seats.

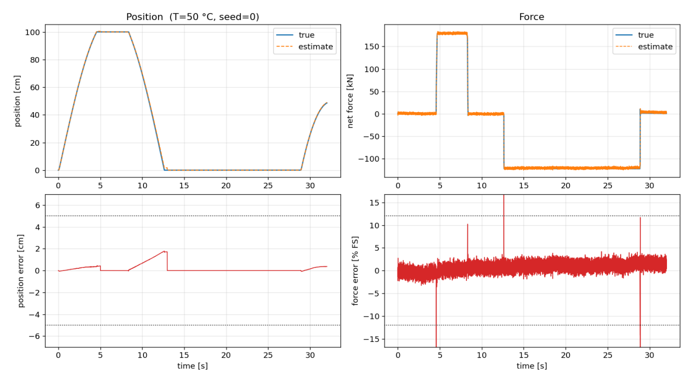

Fig. 10. Representative run at 50 °C. Left: true vs estimated position and the error against the ± 5 cm band (note the bias drift growing between homing events and resetting at each seat). Right: net force and its error against the $\pm 12\,\%$ FS band; the brief spikes are the flow-reversal transients a delayed transducer cannot track. Dotted lines are the limits.

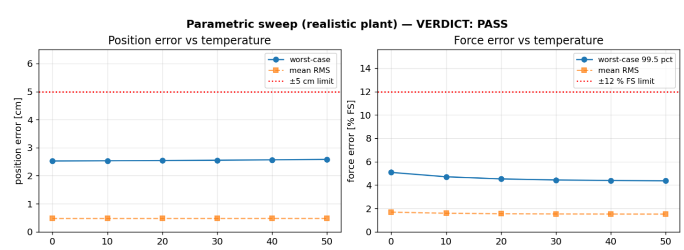

Fig. 11. Worst-case and mean error versus oil temperature. Both metrics stay flat and far below the limits across 0–50 °C.

D. Hose length: how far can the sensors sit from the cylinder?

A real installation places the transducers at the valve manifold and runs a hose to the cylinder (Sec. III), so the sensed pressure and flow are *upstream* of a line loss. We sweep the hose length from 0.5 to 25 m (standard 3/4 in. mobile-hydraulics pressure hose) at three oil temperatures and ask how this degrades each estimate (Fig. 20).

The two channels respond oppositely, and the reason is instructive. **Position is almost length-independent** (flat at $\sim$2.9 cm out to 25 m): the friction drop along the hose does not destroy flow—in steady state the flow that enters the hose at the manifold is the flow that reaches the cylinder—so integrating the measured flow still recovers the stroke, and the only residual is the small transient stored in the line compliance. **Force degrades with length**, because force is read straight off the manifold pressure, which exceeds the true cylinder pressure by the line drop $\Delta p = Q/K_c$. That drop grows with length and, through the viscosity in $K_c \propto 1/\mu$, with *cold* oil: at 50 °C the force error stays near 5 % FS even at 25 m, but at 0 °C it crosses the $\pm 12\,\%$ FS limit at $\sim$15 m. The worst case is therefore a long hose at cold start—a concrete, physically-grounded design rule: *position tolerates an arbitrarily distant manifold, but the force specification sets*

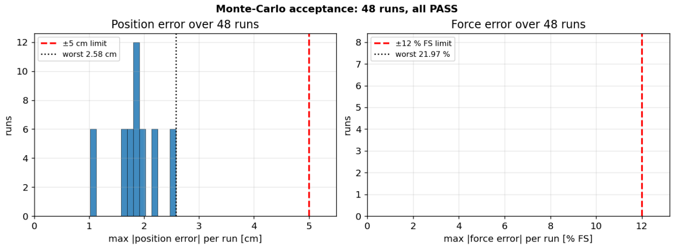

Fig. 12. Error distributions over all 48 Monte-Carlo runs. Every run is inside both acceptance limits (red dashed); worst cases marked.

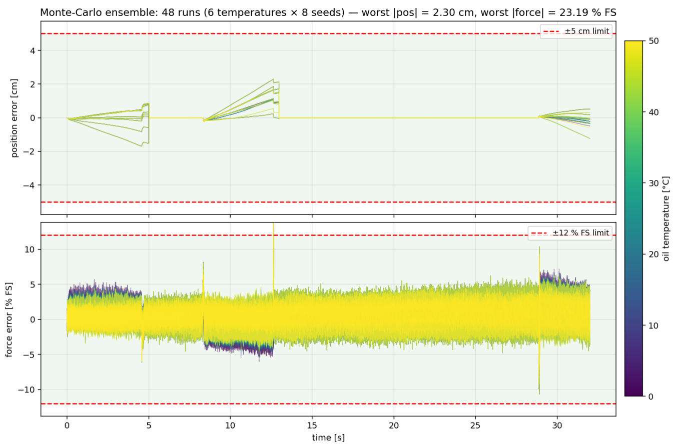

Fig. 13. Monte-Carlo ensemble: all 48 error-vs-time traces overlaid, coloured by oil temperature, against the acceptance bands (green shading, red dashed limits). Top: position error; bottom: net-force error. The position spread stays inside ± 5 cm with the largest excursions mid-stroke between homing events; the force band sits near ± 4 % FS.

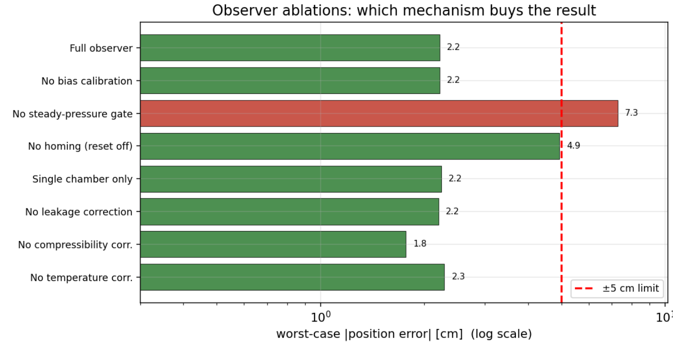

Fig. 14. Observer ablations (worst case over a temperature/seed subset, log scale). Removing the steady-pressure gate or the homing reset is what breaks the result; the physical corrections are minor.

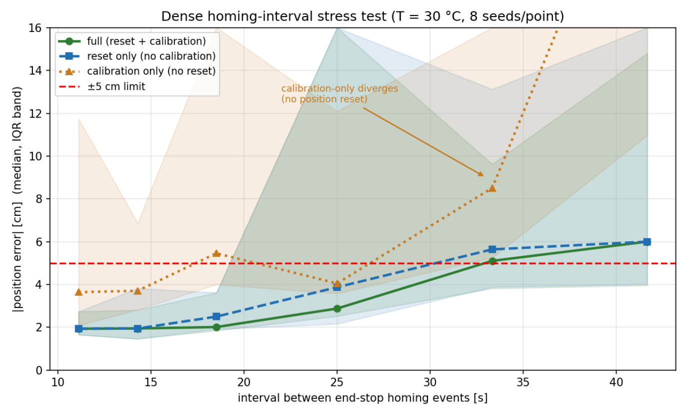

Fig. 15. Dense homing-interval stress test (median, inter-quartile band; 8 seeds). Full (●) and reset-only (■) nearly coincide and hold ± 5 cm out to ~ 30 s; calibration-only (▲, no position reset) is worse everywhere and diverges. The homing reset, not the bias calibration, is what bounds the error.

a maximum hose length that shortens as the oil gets colder. The fix, if a longer run is unavoidable, is to subtract a modelled $\Delta p(Q, T)$ from the measured pressure—the same line model used here, run forward. Figure 19 shows the mechanism directly in a single Hopsan run: a standing offset between the measured (manifold) and true (cylinder) pressures—the force-channel error—plus a sharp transient at each direction reversal.

E. Open-loop excitation at bench scale

All results so far drive the plant with a position controller. To check that the estimator does not secretly lean on the loop, we replaced the controller with an *open-loop* sinusoidal spool command (a separate Hopsan model variant in which the valve input is wired directly to the signal generator rather than to the PI controller) and re-scaled the rig to a small laboratory cylinder: 40 mm bore, 16 mm rod, 0.1 m stroke, 3 kg load, 100 bar supply, with the flow meter and transducer sized to the bench (20 L/min, 120 bar). A benchtop rig plumbs the cylinder directly at the valve block, so unlike the manifold-sensed industrial case (Sec. VII-D) the sensors sit at the cylinder and this variant is line-free. Because the stroke is now 0.1 m, the ± 5 cm position target is expressed as its geometric equivalent, ± 5 % of stroke. Across 0–50 °C and eight mismatch seeds the estimator holds 4.5 % **of stroke worst-case position** and 4.9 % **FS force** (Fig. 21), meeting both targets without a controller in the loop.

Reaching that number required fixing a failure mode that only *open-loop* excitation exposes, and which is instructive about what the homing reset actually depends on. The reset must decide *which* stop a seated piston rests against. Under closed-loop control the answer is trivial: the controller pushes the piston into the commanded stop, so the pressurized chamber names the stop. Under open-loop excitation it is not, because near a stop the spool sine keeps cycling and both chambers can dead-head close to supply pressure. A naive "chamber with the higher pressure wins" rule then mis-identifies the stop on a few-bar difference, and— since the cap area exceeds the rod area—a force-balance rule ($P_1 A_1 > P_2 A_2$) is no better, biasing every dead-headed seat toward "extended." The pressures simply do not carry the stop identity when both chambers are pressurized. The robust, still-sensorless cue is *direction of travel*: the piston rests against whichever stop it last moved toward. We latch the sign of the displacement accumulated over each off-stop excursion (committing it only past 30 % of stroke, so dwell jitter— which can flip the instantaneous velocity sign, the more so

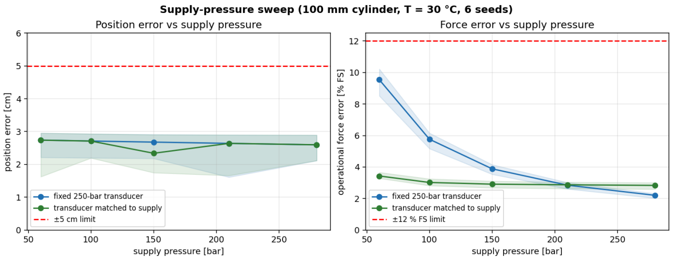

Fig. 16. Supply-pressure sweep (100 mm cylinder). Position is flat; force error rises as $1/P$ with a fixed 250-bar transducer but is flat with a transducer matched to the supply.

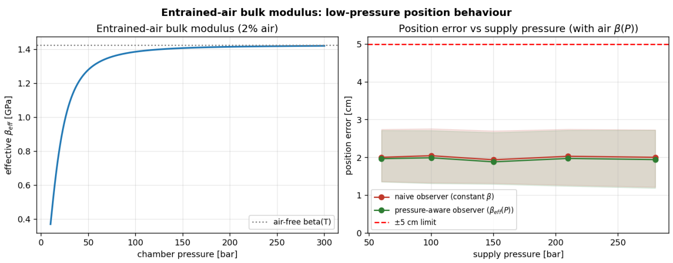

Fig. 17. Entrained-air bulk modulus. Left: $\beta_{eff}(P)$ collapses below $\sim$ 50 bar. Right: yet position error stays flat and a naive constant-β observer matches a pressure-aware one, because the flow-based position estimate is β-insensitive.

at 50 °C where leakage corrupts the velocity estimate—cannot flip the latch). With the travel-direction cue the divergent seeds vanish at every temperature. The closed-loop results above are unchanged by this modification (direction of travel and pressurized chamber agree when a controller holds the stop), so it is a strict generalization. The lesson reinforces Sec. VII-A: homing is load-bearing, but only if the reset can reliably tell the stops apart—and outside closed-loop control that identification must come from motion history, not instantaneous pressure.

F. Comparison with a well-scaled EKF

We implemented the Extended Kalman Filter of Appendix C as a direct competitor, with the engineering needed to make it run on this stiff plant: non-dimensionalized states (fixing the conditioning that made an earlier prototype diverge), semi-implicit integration for the fast pressure mode, both pressures *and* both flows as measurements, an end-stop force in the model, and seated pseudo-measurements (x=stop, v=0) that are the filter's analog of homing. It is now numerically stable and converges. Table III and Fig. 22 give the head-to-head on identical truth and measurements.

The force estimates are comparable (the EKF's is marginally better, coming from its filtered pressure state). On position, though, the observer is clearly better and more robust: the EKF lags the first stroke until its first homing and shows larger mid-stroke and reversal excursions for unfavorable mismatch draws, and its 2σ band is optimistic (82 % coverage versus the ideal 95 %). The observer's algebraic velocity and held-bias structure are simply better matched to a plant where

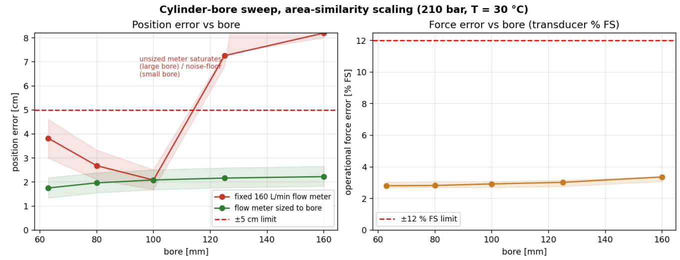

Fig. 18. Bore sweep (63–160 mm, area-similarity scaling). With a flow meter sized to the cylinder, position and force are flat across sizes; a fixed 160 L/min meter saturates at large bores and is noise-floor-limited at small.

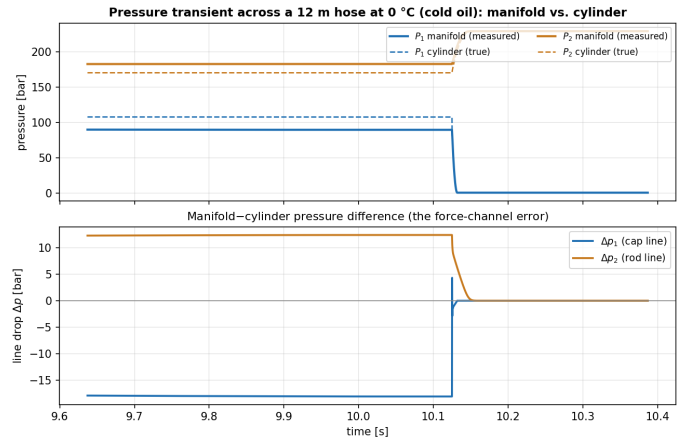

Fig. 19. Hopsan pressure transient across a 12 m hose at 0 °C. Top: measured manifold pressures (solid) sit $\sim$ 12–18 bar off the true cylinder pressures (dashed)—a force read directly from the manifold inherits this gap. Bottom: the line drop Δp for each chamber, steady between events and spiking at the reversal near 10.1 s.

absolute position is weakly observable and the dynamics are stiff. The EKF's one real advantage is the covariance it carries (the shaded band in Fig. 22); making that both tight and well-calibrated would need further tuning. This is consistent with the ablation: the homing reset bounds position, and both estimators depend on it.

G. External validation against a physical rig

Our results are simulation-based by construction, so it matters that an independent *experimental* study reaches a compatible conclusion on real hardware. Zhou et al. [5] reconstruct the position of a fully-mechanized mining support cylinder (180 mm bore, 120 mm rod, 900 mm stroke, 500 kg, water-glycol emulsion) by the same flow-integration principle, predicting flow from the inlet/outlet pressures through a throttle law and integrating Q/A. On a 400 L/min test platform they sweep the supply from 5 to 31 MPa and report the relative error between predicted and measured flow (their Table 8); since position is the time integral of Q/A, that flow-coincidence error is, to first order, their position error.

We reconfigured our Hopsan plant and observer to their rig—geometry, load, emulsion bulk modulus (2.01 GPa), Coulomb/static friction (7/10.5 kN), a 0–40 MPa transducer and a 400 L/min flow meter—and swept the same supply

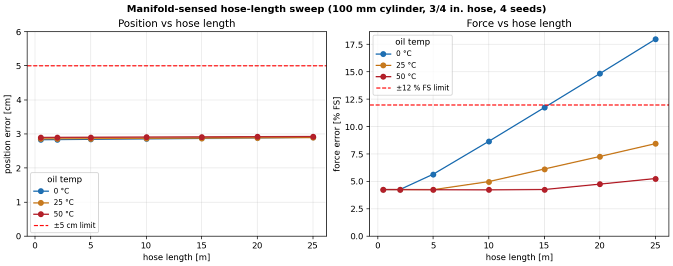

Fig. 20. Hose-length sweep with manifold-located sensors (100 mm cylinder, 3/4 in. hose). Position (left) is nearly length-independent because flow is conserved along the hose; force (right) carries the full line-loss offset, which grows with length and with cold oil (higher viscosity), crossing ±12 % FS near 15 m at 0 °C.

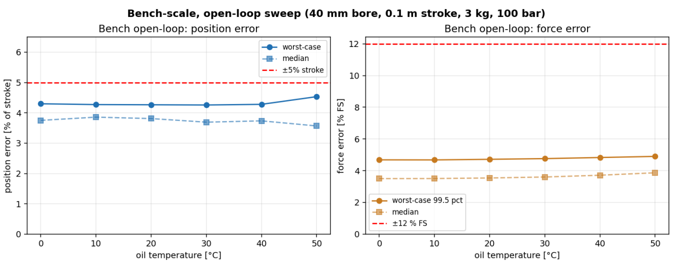

Fig. 21. Bench-scale, *open-loop* sweep (40 mm bore, 0.1 m stroke, 3 kg, 100 bar; no position controller). Worst-case position stays at $\sim 4.3\,\%$ of stroke and force below 5 % FS across 0–50 °C, once the homing reset identifies the seated stop from direction of travel rather than from chamber pressure.

pressures, with the plant bulk modulus following the entrained-air $\beta_{\text{eff}}(P)$ model so that the real low-pressure softening they invoke is present. The driven stroke reaches 296 L/min peak flow, matching their measured range. Figure 23 and Table IV compare the two.

Two things follow. **At and above** 15 **MPa** the experiment and our observer agree closely—both sit below 5 %, ours at 2.4–3.2 % of stroke, well inside their experimental band—so the core mechanism is corroborated on real hardware at the pressures mining supports actually run. **Below** 10 **MPa** the comparison is more interesting: Zhou et al. degrade to 12–20 %, and attribute it explicitly to the bulk modulus varying in the low-pressure region ("the elastic modulus is equivalent to a constant when system pressure is above 10 MPa"). That is precisely the entrained-air $\beta_{\text{eff}}(P)$ physics of Sec. VII (Fig. 17). Our observer, however, stays flat at $\sim 2.4\,\%$ there, because it carries the compressibility term $(V/\beta)\dot{P}$ explicitly and re-zeros at end-stops, whereas a steady-state pressure-to-flow fit cannot separate compression flow from motion flow when the fluid is soft. The experiment thus both validates our high-pressure numbers and independently confirms the low-pressure mechanism we model—while highlighting why a compressibility-aware, homing observer is more robust there than a steady-state flow fit. The residual experimental scatter at 5–10 MPa also reflects flow-meter response lag at low flow, a sensor effect orthogonal to the estimator. (Force is not reported in [5]; our force error rises at low supply only through fixed-transducer SNR, the $1/P$ effect of Fig. 16, reaching $\sim 22\,\%$ FS at 5 MPa with a 40 MPa transducer and falling below 5 % once the supply exceeds 20 MPa.)

TABLE III
EKF VS COMPLEMENTARY OBSERVER (WORST CASE OVER 24 RUNS)

Metric	Observer	EKF
Position error, full run (cm)	2.58	21.5
Position error, post-convergence (cm)	2.58	3.9
Position RMS (cm)	0.65	3.6
Force error, 99.5 pct (% FS)	5.08	4.90
Covariance calibration (within 2σ)	—	82 %

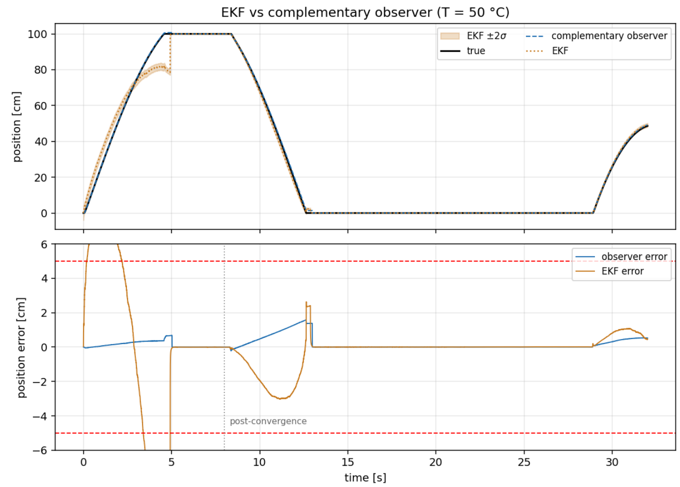

Fig. 22. EKF vs complementary observer on one run (50 °C). Top: position, with the EKF $\pm 2\sigma$ band. Bottom: errors against the ±5 cm band. The EKF lags the first ramp until its first homing and has larger transients; the observer stays tighter throughout.

VIII. Discussion

The result rests on one mechanism. Position from flow integration is *bias-limited*, not noise-limited: zero-mean flow noise integrates to a small random walk, but a constant flow-meter offset integrates to unbounded drift. Early in the study, runs with unlucky bias draws produced 13 cm of drift during a long mid-stroke excursion. The fix was not a better filter but recognising that the bias is directly observable wherever the piston seats a hard stop, and gating that calibration on steady pressure so transient seats do not corrupt it. The ablation (Fig. 14) confirms this: the gate and the homing reset are what hold the result; the leakage/compressibility/temperature corrections are refinements. With realistic per-run parameter mismatch the residual position error rises only from ~ 2.1 to ~ 2.9 cm, so the scheme tolerates the 10–20 % fluid-parameter uncertainty a field estimator would face.

Force, being a direct pressure read-out, is accurate in operation (99.5th-percentile $\sim 4\,\%$ FS) but is *transducer-limited at fast reversals*: when a chamber drains toward cavitation in under a sample, a delayed bandwidth-limited sensor reports a stale value and the instantaneous force is briefly wrong. This is a real measurement limit, and a force-from-pressure scheme inherits it.

TABLE IV
POSITION-ESTIMATION ERROR VS. THE EXPERIMENT OF [5]

Supply [MPa]	Zhou et al. exp. (flow err., range)	This work (pos., median)	This work (pos., worst)
5	12.1–20.1 %	2.4 %	3.0 %
10	0.0–13.8 %	2.4 %	3.1 %
15	0.0–0.5 %	2.4 %	3.1 %
20	0.5–3.2 %	2.4 %	3.1 %
25	0.5–4.8 %	2.4 %	3.1 %
30	1.4–4.9 %	2.3 %	3.0 %

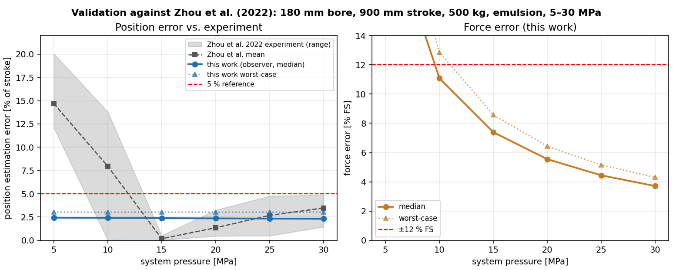

Fig. 23. Validation against the physical experiment of [5] (180 mm bore, 900 mm stroke, 500 kg, emulsion). Left: our observer's position error (blue) against their measured flow-coincidence error envelope (grey). The two agree below 5 % above 15 MPa; below 10 MPa the experiment degrades with the softening bulk modulus while the compressibility-aware, homing observer stays flat. Right: our force error, which rises at low supply only through fixed- transducer SNR.

The practical reading is that for this class of actuator, periodic end-stop homing converts a hard absolute-position problem into a bounded-drift problem that cheap pressure and flow sensing can solve to a few centimetres. The stress test (Fig. 15) makes the dependence quantitative: the budget holds only while the actuator seats a stop often enough. A process that parks mid-stroke for long periods would drift past target and would need an independent absolute reference.

IX. LIMITATIONS

This is a simulation spike, not a hardware validation. The plant now includes Coulomb/stiction friction, finite valve bandwidth and per-run parameter mismatch, and the sensors include bandwidth, delay, drift and saturation; but air entrainment, valve hysteresis/deadband, structural compliance and sensor failure modes are still simplified or absent, and the TLM solver is a validated model, not measured hardware. Temperature is a steady per-run constant; within-stroke thermal transients are out of scope. The envelope now spans 60–280 bar and 63–160 mm bore, but still one duty-cycle family that periodically seats stops. The force result remains, fundamentally, pressure sensing rather than a hard inference problem, and its accuracy degrades at fast pressure reversals.

X. CONCLUSION

Within a 0–50 °C envelope, with realistic sensor non-idealities, Coulomb friction, valve dynamics and 10–20 % plant/estimator parameter mismatch, a double-acting hydraulic cylinder modelled in Hopsan can be position-estimated to 2.58 cm and force-estimated to 5.08 % of full scale (operationally) without a position sensor, inside the ±5 cm and ±12 % targets, using flow-continuity velocity with end-stop bias calibration and pressure-derived force. A three-mode stress test identifies the end-stop *homing reset* as the load-bearing mechanism (it zeros error from all sources at each seat); bias calibration is a secondary refinement that cannot bound drift alone, and the ±5 cm budget yields a concrete homing-interval design rule. The harness, with Hopsan as the truth plant feeding a sensor and observer pipeline, is reusable for sharper questions: drift between homing events, extended Kalman variants with covariance bounds, and a hardware bridge through Hopsan's FMU export. Detailed equations, parameter tables, and the full human/AI session log are in the appendix.

ACKNOWLEDGEMENTS

This work, including the simulation harness, the Hopsan integration and the writing, was carried out with the assistance of Anthropic's Claude (Claude Code), used both for running the experiments and for drafting this paper. All design decisions, target setting and verification were directed by the author.

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APPENDIX A
FULL PARAMETER SET

Table V lists every parameter, complementing the abbreviated Table I. The temperature curves are $\beta_e(T) = \beta_0\,[1 + k_\beta(T - T_0)]$ with $\beta_0 = 1.5$ GPa, $T_0 = 20$, $k_\beta = -0.005\,°\mathrm{C}^{-1}$ (clamped at 0.3 GPa); $\mu(T) = \mu_{40} \exp\big(B(1/T_K - 1/T_{K,40})\big)$ (Vogel-like) with $\mu_{40} = 41$ mPa s and $B = 3100$ K; and $C_{\text{leak}}(T) = C_0\,\mu_{40}/\mu(T)$ with $C_0 = 2 \times 10^{-13}$ m^3/(s Pa).

TABLE V
COMPLETE PARAMETER SET

Symbol	Value	Meaning
A_1	$7.854 \times 10^{-3}\,\mathrm{m}^2$	cap-side area
A_2	$5.391 \times 10^{-3}\,\mathrm{m}^2$	rod-side area
s_ℓ	$1.0\,\mathrm{m}$	stroke
V_{01}, V_{02}	$2 \times 10^{-4}\,\mathrm{m}^3$	dead volumes
m	200 kg	moving mass
P_s	210 bar	supply (relief-set)
F_{FS}	164.9 kN	force full scale
β_0	1.5 GPa	bulk modulus @ 20 °C
μ_{40}	41 mPa s	viscosity @ 40 °C
C_0	2×10^{-13}	leak coeff @ 40 °C
C_q	0.67	valve discharge coeff
ref. freq / amp	0.04 Hz / 1.1 s_ℓ	sine reference
K_p, K_i	0.03, 0.004	PI gains (spool m / m)
sample rate	1 kHz	sensor + observer
duration	32 s	per run

APPENDIX B
OBSERVER ALGORITHM

The observer runs at the sample rate. Per step:

1) Low-pass the measured pressures (40 Hz) and flows (25 Hz); form $\dot{P}_i$ by finite difference of the filtered pressure.
2) Evaluate β_e, C_{leak} from the noisy temperature reading.
3) Compute per-chamber velocities v_1, v_2 from the rearranged continuity relations; blend by flow magnitude with a small regulariser so $v \to 0$ when both flows vanish.
4) Integrate $x \mathrel{+}= v\,\Delta t$, clamped to $[-0.02,\ s_\ell{+}0.02]$.
5) If seated (filtered flows small and $\max(P_1, P_2) > 0.7\,P_s$) for > 0.3 s: pin x to the corresponding stop; if additionally $\dot{P} \approx 0$ (steady), update $b_1 = Q_1 - q_{\text{leak}}$, $b_2 = Q_2 + q_{\text{leak}}$.
6) Output $\hat{x}$ and $\hat{F} = P_1A_1 - P_2A_2$.

The steady-pressure gate in step 5 is the single most important detail: without it, transient seatings (arrive-and-immediately-leave) capture compressibility flow as if it were bias and corrupt the calibration.

APPENDIX C
STATE-SPACE MODEL AND THE EXTENDED KALMAN FILTER

This section gives the principled estimator the observer of Sec. V approximates: a nonlinear state-space model and the Extended Kalman Filter (EKF) that is its textbook recursive estimator. We then explain why the deployed estimator is the reduced complementary observer rather than the full EKF.

A. Nonlinear state-space model

Take the state, augmented with the two flow-meter biases and the external load so they are estimated rather than assumed,

$$\boldsymbol{\xi} = \big[\, x,\ v,\ P_1,\ P_2,\ b_1,\ b_2,\ F_L \,\big]^\top. \tag{5}$$

With measured flows Q_1, Q_2 and temperature as inputs, the continuous dynamics $\dot{\boldsymbol{\xi}} = \mathbf{f}(\boldsymbol{\xi}, \mathbf{u})$ are

$$\dot{x} = v, \tag{6}$$
$$\dot{v} = \big(P_1A_1 - P_2A_2 - F_{\text{fric}}(v) - F_L\big)/m, \tag{7}$$
$$\dot{P}_1 = \tfrac{\beta_e}{V_1(x)}\big[(Q_1 - b_1) - A_1 v - q_{\text{leak}}\big], \tag{8}$$
$$\dot{P}_2 = \tfrac{\beta_e}{V_2(x)}\big[(Q_2 - b_2) + A_2 v + q_{\text{leak}}\big], \tag{9}$$
$$\dot{b}_1 = \dot{b}_2 = \dot{F}_L = 0 \quad \text{(random-walk states)}, \tag{10}$$

with $V_1(x) = V_{01} + A_1 x$, $V_2(x) = V_{02} + A_2(s_\ell - x)$ and $q_{\text{leak}} = C_{\text{leak}}(P_1 - P_2)$. Only the two chamber pressures are measured,

$$\mathbf{z} = \mathbf{h}(\boldsymbol{\xi}) + \boldsymbol{\nu} = [\,P_1,\ P_2\,]^\top + \boldsymbol{\nu}. \tag{11}$$

Position is observable because x enters the pressure dynamics through the volumes $V_i(x)$: $\partial\dot{P}_i/\partial x \neq 0$ whenever there is net chamber flow, so a pressure innovation carries information about x.

B. EKF recursion

Discretising to $\boldsymbol{\xi}_{k+1} = \mathbf{f}_d(\boldsymbol{\xi}_k, \mathbf{u}_k) + \mathbf{w}_k$ with $\mathbf{w}_k \sim \mathcal{N}(0, \mathbf{Q})$ and $\mathbf{v}_k \sim \mathcal{N}(0, \mathbf{R})$, the EKF alternates

$$\text{predict:}\quad \hat{\boldsymbol{\xi}}^- = \mathbf{f}_d(\hat{\boldsymbol{\xi}}, \mathbf{u}), \quad \mathbf{P}^- = \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q}, \tag{12}$$
$$\text{update:}\quad \mathbf{y} = \mathbf{z} - \mathbf{h}(\hat{\boldsymbol{\xi}}^-), \quad \mathbf{S} = \mathbf{H}\mathbf{P}^-\mathbf{H}^\top + \mathbf{R}, \tag{13}$$
$$\mathbf{K} = \mathbf{P}^-\mathbf{H}^\top\mathbf{S}^{-1}, \quad \hat{\boldsymbol{\xi}} = \hat{\boldsymbol{\xi}}^- + \mathbf{K}\mathbf{y}, \tag{14}$$
$$\mathbf{P} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}^-(\mathbf{I} - \mathbf{K}\mathbf{H})^\top + \mathbf{K}\mathbf{R}\mathbf{K}^\top, \tag{15}$$

the last in Joseph form for numerical stability. Here $\mathbf{F} = \partial\mathbf{f}_d/\partial\boldsymbol{\xi}|_{\hat{\boldsymbol{\xi}}}$ and $\mathbf{H} = \partial\mathbf{h}/\partial\boldsymbol{\xi} = [\,\mathbf{0}\ \mathbf{I}_2\ \mathbf{0}\,]$ (it selects P_1, P_2). The load-bearing Jacobian entries are the pressure sensitivities

$$\frac{\partial\dot{P}_i}{\partial v} = \mp\frac{\beta_e A_i}{V_i(x)}, \qquad \frac{\partial\dot{P}_i}{\partial x} = -\frac{\beta_e A_i}{V_i(x)^2}\,(\cdots), \tag{16}$$

which couple the pressure measurement to velocity and position, and $\partial\dot{P}_i/\partial b_i = -\beta_e/V_i$, which lets the filter learn the flow-meter biases. $\mathbf{Q}$ and $\mathbf{R}$ are set from the process and sensor-noise magnitudes of Table V.

C. Implementation and conditioning

A direct EKF on this plant is numerically delicate. The hydraulic stiffness makes $\beta_e/V_i \sim 10^{12}$–10^{13}, so the pressure Jacobian entries above are enormous, while the state magnitudes span $\sim 10^7$ (Pa) down to $\sim 10^{-2}$ (m/s) and $\sim 10^{-5}$ (m³/s, bias): the covariance $\mathbf{P}$ is severely ill-conditioned, and an early prototype that inferred velocity by integrating the force balance (2) diverged. We made a runnable EKF with four ingredients: (i) *non-dimensionalisation* – each state is divided by a physical scale (stroke, 0.3 m/s, P_s, a bias scale, F_{FS}) so the scaled covariance is $O(1)$ and Jacobians are taken numerically in scaled space; (ii) *semi-implicit (symplectic) Euler* sub-steps for the prediction, stable for the v–P oscillator where forward Euler at 1 kHz is not; (iii) the two *flows as measurements* ($h_Q = \pm A_i v + b_i + q_{\text{leak}} + (V_i/\beta_e)\dot{P}_i$, the compressibility term taken from the filtered measured pressure) to make velocity and bias observable; and (iv) seated *pseudo-measurements* (x=stop, v=0) as the filter's homing. The complementary observer of Sec. V can then be read as a structurally-reduced form of this EKF: its velocity blend is the flow-measurement update on v, and its gated end-stop calibration is the v=0 pseudo-measurement that separates bias. Sec. VII-F benchmarks the two; the EKF is stable and competitive on force but worse and less robust on position, so the reduced observer is deployed and the EKF is kept for the covariance estimate it provides.

APPENDIX D
HOPSAN INTEGRATION

A. Parameterization

For each run the model parameters (geometry, the temperature-dependent β_e, C_{leak} and viscous friction, the PI gains and the sine reference) are overridden, the model is simulated, and the pressure, flow and position time series are exported to the estimation pipeline.

B. Chamber pressure/flow logging

In Hopsan's TLM, two ports sharing a node are logged on one side, so the cap and rod chamber pressures and flows are read on the valve side (PA↔cap A_1, PB↔rod A_2): pressure is identical either side of a node, and the valve-port flow already equals flow into the chamber. Position is the shared cylinder/mass position.

C. Controller scaling

The valve `in` port is spool position in metres ($|x_v| \leq x_{v\,\max} \approx 0.01$), so $K_p \approx 0.03$ maps a $\sim$1 m position error to a $\sim$0.01 m spool command. An early misconfiguration with $K_p = 5$ slammed the valve fully open, producing an unphysical 2000 L/min and 358 bar; reducing the gain to the example's scale fixed it.

APPENDIX E
PER-TEMPERATURE RESULTS

Table VI gives worst-case (over eight seeds) and mean-RMS errors at each temperature. Both metrics are essentially flat, confirming that the end-stop homing reset neutralises accumulated drift across the envelope.

TABLE VI
WORST-CASE / MEAN-RMS ERROR BY TEMPERATURE (8 SEEDS EACH)

T (°C)	pos max (cm)	pos RMS (cm)	force 99.5 (% FS)	force RMS (% FS)
0	2.76	0.54	4.20	1.58
10	2.77	0.54	4.20	1.58
20	2.79	0.54	4.20	1.58
30	2.81	0.55	4.20	1.58
40	2.83	0.55	4.20	1.57
50	2.85	0.55	4.20	1.57

APPENDIX F
FAILURE MODES ENCOUNTERED

The spike surfaced several issues worth logging, since they shaped the design:

- **Position drift (13 cm).** Long mid-stroke excursions with unlucky bias draws drifted far past target. Root cause: bias calibration was firing on transient seats. Fix: steady-pressure gate plus a duty cycle that dwells at stops.
- **Sign of valve-port flow.** The chamber-inflow convention required no sign flip on `PA/PB#Flow`; an initial flip inverted the velocity (negative correlation with $\dot{x}$), caught by a correlation check.
- **Controller gain.** See D; a unit mismatch drove nonphysical flows until corrected.
- **(Earlier pure-Python plant, since replaced by Hopsan).** An explicit-RK4 prototype suffered end-stop-damping instability and cavitation blow-up, and an EKF that derived velocity by double-integrating the force balance diverged to over a metre of error. These motivated switching to a TLM solver for the truth model and to a flow-continuity observer. They are recorded here as part of the exploration history.

APPENDIX G
FUTURE DIRECTIONS

- Densify the homing-interval / mismatch stress sweep (more seeds and frequencies) and report a tail metric (95th percentile / max-over-seeds) with bootstrap confidence intervals, turning the median/IQR guideline into a tail-reliability design rule.
- Improve the EKF (Sec. VII-F): better-calibrated covariance and faster pre-homing convergence (e.g. a UKF, or modelling the flow-measurement error during transients) to make its confidence band trustworthy.
- Sweep sensor grades to find the cheapest transducers that still pass.
- Bridge to hardware-style co-simulation via Hopsan's FMU export.
- Add within-stroke thermal transients with a custom temperature-aware cylinder component.

APPENDIX H
SESSION FLOW: HUMAN VS. AI EFFORT

Per the research-process requirement to preserve what was human input and what was AI effort, the spike proceeded through these human prompts (paraphrased), each answered by AI analysis, code and verification:

1) *"A double-acting cylinder is coupled because flow and pressure at both ends set position and force, correct?"* AI: clarified the coupling (flow→velocity→position; pressure→force) and offered a state-space model.
2) *"Set up a parametric offline simulation showing position error $\leq \pm 5$ cm and force $\leq \pm 12$ % FS across full stroke and 0–50 °C with realistic noise."* AI: built a first plant, sensors, observer and sweep; iterated through stability and drift bugs to a passing result.

3) *"Can we use Hopsan for the modelling and simulation?"* AI: researched Hopsan's component library, TLM solver, temperature handling and interfaces; recommended Hopsan as the plant with the estimator kept in Python.
4) *"Refactor to only use Hopsan as the plant."* AI: vendored the Position-Servo example, swapped the reference to a sine source, set up the parameter overrides, and validated the integration.
5) *"Run the sweep on the real simulator."* AI: debugged the chamber pressure/flow logging and controller scaling, and ran the full sweep.
6) *"Treat this chat as a research spike and write it up in IEEE format, with figures of the sim, Hopsan, and plots."* AI: built (from source) and captured the Hopsan model canvas (Fig. 5), generated the figures and this paper, and ran an external (Codex) review.
7) *"Make the simulation more realistic."* AI: added per-run plant/ estimator parameter mismatch, Coulomb/stiction friction, finite valve bandwidth and full transducer dynamics (bandwidth, delay, drift, saturation); added the ablation study and the homing-interval stress test, which together identified the homing reset as the load-bearing mechanism.
8) *"Add the EKF theory, then implement a well-scaled EKF and benchmark it."* AI: derived the state-space model and EKF recursion (App. C), implemented a non-dimensionalised, semi-implicit filter, and benchmarked it head-to-head against the observer (Sec. VII-F).
9) *"Compare with the Simscape lab model; is TLM any different?"* AI: contrasted the acausal-DAE and transmission-line approaches and added the solver-rationale discussion (Sec. III-A).
10) *"Make an open-loop variant and extend the sweep to bench scale."* AI: built the open-loop model variant, found and fixed an open-loop-only stop-identification failure (direction-of-travel homing), and validated the bench case (Sec. VII-E).
11) *"Validate against the Zhou et al. experiment."* AI: reconfigured the plant to their physical rig and showed agreement above $15\,\mathrm{MPa}$ plus independent confirmation of the low-pressure bulk-modulus mechanism (Sec. VII-G).
12) *"Add hose line losses with manifold-located sensors; sweep hose length."* AI: added the volume+orifice line model, moved ground-truth force to the cylinder side, and produced the hose-length design rule (Sec. VII-D).
13) *"Make it readable from first principles and more visual."* AI: added the principle, observer and manifold diagrams (Figs. 1, 8, 2), a plain-language primer and roadmap, and tightened the abstract. Final sweep: worst position $2.58\,\mathrm{cm}$, operational force $5.08\,\%$ FS.

All acceptance targets, the choice of Hopsan, and the scope were set by the human; the modelling, coding, debugging, figure generation and drafting were AI effort under that direction.

Appendix I
Code and Data Availability

The harness (`cylinder_sim/`: parameters, Hopsan driver, sensor model, observer, sweep, plots), the Hopsan model (`hopsan/sensorless_cylinder_plant.hmf`), the integration tests, and the figure-generation scripts accompany this paper in the spike directory.